

COURSE PROJECT

Project Proposal

1 Title & Team

Give the project a clear, specific name. List everyone on the team and how to reach them.

Project Title	Synthesis: A Voice-Enabled, Retrieval-Grounded Interview-Prep Platform
Team Name	Cynthèse
Submission Date	12 June 2026

Team Member	Email	Program / Year
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2 Executive Summary

Half a page, written last. A reader who stops here should still grasp the whole project: what problem you tackle, what you will do, and what success looks like.

- Synthesis is a voice-enabled platform that prepares a candidate for one specific job: it produces a measurable fit-to-field score between a resume and a posting, rehearses the behavioural round against the candidates own answers and standard behavioural questions, and runs adaptive, retrieval-grounded case interviews.
- It matters to every student and early-career applicant facing interviews. Case and behavioural prep today is expensive, generic, and gives no objective feedback; Synthesis makes it personal, on-demand, and grounded in real material — the candidate's own resume and live job postings.
- The outcome is a working three-module platform plus a measured result: a fit scorer validated on 2,484 labelled resumes across 24 job families, reported as top 1 and top 3 category match accuracy with a confusion matrix and an embeddings-versus-structured ablation.

3 Problem Statement

Define the problem precisely. Anchor it in evidence rather than assumption and make clear why it is worth solving now.

- Interview preparation — both case and behavioural — is hard to rehearse realistically and even harder to get objective feedback on. A candidate cannot easily tell whether they are a strong fit for a given role, nor whether their answers land with an interviewer. The feedback loop that would make practice productive simply does not exist for most applicants.
- Students and early-career applicants feel this most acutely. Weak preparation costs interviews, offers, and time, yet the tools available to them are generic, expensive, or give no measurable signal: paid mock-interview services are costly and inconsistent, and free resources cannot tell a candidate how their profile compares to the requirements of a specific posting.
- The opportunity is concrete and measurable. A labelled corpus of 2,484 real resumes across 24 job families, a one-month snapshot of 123,849 live LinkedIn postings with their skills and salary side-tables, and O*NET 30.3 — the US Department of Labor's taxonomy of 1,016 occupations — together let us demonstrate that fit-to-field can be scored objectively. The resume labels give us ground truth: we can verify whether a resume scores highest against postings from its own field, something current prep tools do not even attempt.
- Why now: LLMs have made reliable structured extraction from messy resumes and job descriptions practical for the first time, and all three datasets are public and already in hand — the pieces to build and validate an objective fit signal are available today.

4 Objectives & Scope

State the concrete objectives, then draw the boundary. Being explicit about what is out of scope is as important as what is in.

Objectives	Build and deploy a working interview-prep platform with three modules — a resume-to-JD fit analyzer, a behavioural simulator, and a case simulator — usable end-to-end by the July 30 demo. Engineer a document parser robust to messy real-world resumes and job descriptions, returning valid structured output on at least 95% of a held-out test sample. Validate the fit scorer on the 2,484-resume labelled corpus, achieving top-1 category-match accuracy of 70% or higher (top-3 of 90% or higher), reported with a confusion matrix. Ground every interview question and fit judgement in retrieved evidence from O*NET and live postings, scored against a fixed rubric rather than free-form model opinion.
In Scope	The fit analyzer with a validated, interpretable score; the case and behavioural simulators in text, with browser-based voice or text on the behavioural module; a retrieval (RAG) layer over O*NET and the postings corpus with an offline ingestion pipeline; and structured, evidence-backed scoring.
Out of Scope	A native mobile app; multi-tenant production hardening; training or fine-tuning our own models; and real-time speech-to-speech voice. We deliberately rely on hosted LLMs and pre-trained embeddings rather than training from scratch, to keep the seven-week scope achievable.

5 Data Sources

List every dataset or source you intend to use. Note where it comes from, its format, and whether access is already secured.

Source	Type / Format	Access & Availability	Notes
Resume Dataset (Kaggle, snehaanbawal) kaggle.com/datasets/snehaanbawal/resume-dataset	CSV (text, HTML, PDF)	Public — have it	2,484 resumes, 4 cols, 0 missing; key fields Resume_str, Resume_html, Category (24 families). Quality: clean (3 rows to drop); leakage traps measured — 'Company Name' 100%, ALL-CAPS title 99.5% — both stripped.
LinkedIn Job Postings 2023-24 (Kaggle, arshkon) kaggle.com/datasets/arshkon/linkedin-job-postings	CSV (linked tables)	Public — have it	123,849 postings, 11 linked files; key fields title, description, location (near-complete), skill tags (98.6%). Quality: ~71% lack salary; single 4-week scrape (cross-sectional); 15.3% reposts to drop; cite, don't redistribute.
O*NET (US Dept. of Labor) onetcenter.org/database.html	Database / CSV	Public — have it	Release 30.3; 1,016 occupations (894 with Essential Skills), 45 files. Key fields Essential Skills (IM/LV scales), Tasks, Software Skills. Quality: cleanly licensed; drop Suppress/Not-Relevant rows (Knowledge 8.3%, others <2%).

6 Methodology

Explain how you will get from data to results. Be concrete about methods, tools, and the logic that connects them to your objectives.

- An LLM-driven extraction-and-matching pipeline, with no model training in production. Each job description and resume is parsed into structured requirements and evidence, grounded in the

O*NET taxonomy of 1,016 occupations; requirements are then matched one by one using semantic embeddings plus deterministic rules, and aggregated into a single interpretable fit score with the supporting evidence attached. We choose extraction-plus-matching over an end-to-end classifier precisely because it is auditable: every point in the score traces back to a specific requirement and a specific line of the resume.

- The interview agents run as a finite-state machine with retrieval-grounded questioning and rubric-based scoring, so questions adapt to the candidate while staying anchored to retrieved exemplars rather than improvised by the model. Tools: Next.js on Vercel; Supabase (Postgres + pgvector) for storage and vector search; local sentence-transformer embeddings; the Anthropic Claude API (Haiku for high-volume calls, Sonnet for the demo); n8n for offline ingestion; and the browser Web Speech API for voice.
- Data preparation follows the EDA: drop the three under-populated resume categories and three corrupt rows, normalize full-width dashes, and strip the ALL-CAPS title line so the model cannot read the answer off the first line; on the postings side, de-duplicate the 15.3% reposts and annualize salaries before comparison.
- Validation leans on the labels we already have. Because every resume is tagged with its true job family, we test whether it scores highest against postings from its own field, reporting top-1 and top-3 accuracy, a full confusion matrix, and an ablation that isolates the contribution of embeddings versus structured rules. This turns an otherwise subjective “good fit” claim into a number we can defend.

7 Deliverables

Name the tangible outputs the project will produce, and the form each will take.

#	Deliverable	Description	Format
1	Synthesis platform	Deployed web app with the three modules (fit analyzer, behavioural, case), retrieval, and scoring	Web app (Vercel)
2	Fit-analyzer validation study	Top-1/top-3 accuracy, confusion matrix, and ablation on the labelled resume corpus	Notebook + figures
3	Final presentation & write-up	Methodology, results, and a live demo	Slides / PDF + demo

8 Timeline / Gantt

Lay out the schedule by phase across the available weeks. Shade the cells that each task spans. The bars below are examples — replace them with your own plan.

Phase / Task	W1	W2	W3	W4	W5	W6	W7	W8
Setup, proposal & schemas								
Fit analyzer + validation								
RAG ingestion + retrieval								
Case + behavioural (text)								
Scoring reports & dashboard								
Voice, polish & presentation								

9 Risks

Identify what could derail the project and how you will respond. Rate likelihood and impact as Low / Medium / High.

Risk	Likelihood	Impact	Mitigation
Voice recognition errors (STT mishears finance terms, e.g. EBITDA)	M	M	Demo on Chrome, keep a text fallback, add voice to the behavioural module first
LLM scoring is subjective / inconsistent	M	M	Fixed rubric, retrieved exemplars, low temperature, human spot-checks

Risk	Likelihood	Impact	Mitigation
Compressed seven-week timeline / scope creep	M	H	Phase gates; pre-committed kill-switch (text demo if voice is unstable); build the core first
Data licensing & personal-data privacy	L	M	Cite, don't redistribute scraped data; lean on O*NET; store resumes behind auth + row-level security with delete

10 Roles

Map each team member to what they own. Clear ownership prevents work from falling through the cracks.

Team Member	Primary Responsibilities	Backup For
Feroz (Lead)	Agents & retrieval core: case + behavioural interviewers, coach/evaluator, JD parser, state machine, retrieval wiring; repo owner, schemas, standing-instructions file, PR review	Data / RAG
Rui	Data engineering: ingestion (DOCX/PDF parsing), Supabase + pgvector, embeddings, n8n; data exploration + fit-scorer validation script	Fit analyzer (data side)
Emmanuel	Fit analyzer module (requirement extraction, matching, fit report) + frontend & voice: app UI, upload flow, transcript, scorecards	Content
Ibuen	Content & communications: behavioural answer documents, question bank, case-content & dataset sourcing, slide deck, write-up, user testing	Presentation & testing support

11 Success Metrics

Define how you will know the project succeeded. Each metric should have a target and a clear way to measure it.

Metric	Definition	Target	How Measured
Fit-scorer accuracy	Resume's true field is the top-ranked category	Top-1 $\geq 70\%$ (Top-3 $\geq 90\%$)	Validation script over 2,484 labelled resumes
Parser robustness	Resume and JD parser returns valid structured output	$\geq 95\%$ of test docs parse cleanly	Run over the resume + postings sample
Working demo	All three modules usable end-to-end at the presentation	Fit + behavioural-voice + case-text all run	Live demo on July 30
Running cost	Total API spend across the project	$\leq$ USD 10	Anthropic console usage